The crystal structure of polyethylene terephthalate

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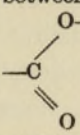
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The crystal structure of polyethylene terephthalate (Terylene) has been determined by the interpretation of the X-ray diffraction patterns given by drawn fibres. The unit cell, which contains one chemical unit, —CO.C_6H_4.CO.O.$(CH_2)_2$O—, is triclinic, with $a = 4{\cdot}56$ Å, $b = 5{\cdot}94$ Å, $c = 10{\cdot}75$ Å, $\alpha = 98\frac{1}{2}°$, $\beta = 118°$, $\gamma = 112°$. This gives the density of the crystals as 1·455 g/cm³.

If it is assumed that the density of amorphous regions in partly crystalline material is the same as that of the entirely amorphous material (1·335), the proportion by weight of crystals in ordinary drawn yarn (density 1·39) is 48 %, and in specimens of the highest recorded density (1·41) it is 64·5 %. These figures are, however, lower limits; if, as is likely, the density of amorphous regions in partly crystalline specimens is lower than 1·335, the proportions of crystalline material are higher than the figures given.

The positions of atoms in the crystals have been deduced from the relative intensities of X-ray diffractions; approximate positions were found by trial-and-error methods, and refinement was then effected by calculations of electron densities, using the three-dimensional Fourier synthesis method. The molecules are nearly planar in configuration. Distances between atoms in neighbouring molecules are all normal Van der Waals contact distances; there is therefore no structural evidence for any unusually strong forces between the molecules. This conclusion is in agreement with that based on an estimate of the cohesion energy, which is very nearly the same as that of a comparable aliphatic polyester, polyethylene adipate.

The high melting-point of polyethylene terephthalate in comparison with aliphatic polyesters is therefore not due to strong forces between the molecules; it is attributed to the rigidity of the aromatic ring with its attached  groups.

In most drawn fibres of this polymer, the crystals do not have their c-axes (those parallel to the chain molecules) exactly parallel to the fibre axes; they are tilted in a precisely defined crystallographic direction, such that the $\bar{2}30$ plane remains vertical and the inclination of the 001 plane to the fibre axis increases. This is especially well defined in fibres heated to 210° C and allowed to relax, the tilt angle being then about 5°. In principle it is expected that polymer crystals of triclinic symmetry would be tilted, but it is not clear what determines the crystallographic direction of tilt. Tilted crystal X-ray diffraction photographs played an important part in the structure determination.

Crystalline aromatic polyesters have sufficiently high melting-points to make them suitable for textile fibres. One of the simplest of these, the polymer made from terephthalic acid and ethylene glycol—polyethylene terephthalate,

$$(—O.CH_2.CH_2.O.CO.C_6H_4.CO—)_p$$

—is now being made on a large scale; its melting-point, 264° C, is almost exactly the same as that of nylon 6:6 (265° C), and drawn fibres of this substance (known as Terylene in this country and Dacron in the U.S.A.) have high tenacity and

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many other physical properties which make them suitable as textile fibres. In the present investigation, the crystal structure has been determined by interpretation of the X-ray diffraction patterns of the fibres. One of the points of interest is the high melting-point: polyamides appear to owe their high melting-points to hydrogen bonds between the amide groups (Bunn & Garner 1947), but esters are not expected to be linked in this way; the search for evidence for or against strong intermolecular forces of some kind was one of the motives behind this investigation.

Preliminary X-ray work showed that spun filaments (made by extrusion of the melt through small holes) give the diffuse scattering which is characteristic of non-crystalline substances; and the uniform intensity of the diffuse rings shows that there is little if any molecular orientation. Filaments drawn above 70° C, on the other hand, give a sharp highly detailed diffraction pattern, showing that they are highly crystalline and that the crystalline regions are well oriented. Examination of spun filaments under the polarizing microscope reveals a very weak positive birefringence ($\sim 4 \times 10^{-3}$), indicating a small but definite degree of molecular orientation. Drawn filaments have an extremely strong birefringence ($\sim 200 \times 10^{-3}$), indicating a high degree of molecular orientation, with the planes of the benzene rings (the chief source of such high birefringence) approximately parallel to the fibre axis.

X-ray diffraction data and unit cell dimensions

The earliest attempt to determine the crystal structure was made by Mrs M. C. M. Stokes at Welwyn, on the basis of X-ray diffraction photographs taken at the Northwich Laboratory of I.C.I. Alkali Division. The fibre repeat distance, about 10·5 Å, was consistent with the idea that the chains are in extended form, since a fully extended planar chain would have a repeat distance of 10·9 Å. Various unit cells were considered in the attempt to account for the positions of the reflexions, and it was concluded that only a triclinic cell would account for them satisfactorily. At about the same time, an equivalent triclinic cell (representing the same spatial arrangement of lattice points but connected up to form a unit cell in a different way) was suggested by Astbury & Brown (1946), their dimensions being $a = 5{\cdot}54$ Å, $b = 4{\cdot}14$ Å, $c = 10{\cdot}86$ Å, $\alpha = 107°\ 5'$, $\beta = 112°\ 24'$, $\gamma = 92°\ 23'$ (cell 1). Only one molecule runs through this cell.

The relative intensities of the reflexions, calculated for a structure based on this cell, gave some measure of agreement with the observed intensities, but failure to improve the agreement by adjusting atomic positions led to a reconsideration of the unit cell. More detailed diffraction patterns from better oriented specimens showed additional reflexions, among them one on the fourth layer line which could not be accounted for by cell 1; and any modification of that cell to account for this reflexion failed to account for other reflexions. Reconsideration of all the data showed that another quite different triclinic cell (corresponding to the first only for the [001] projection) would account for all the 51 reflexions on the best diffraction photographs then available. More recently, still better photographs (of specimens drawn at 75° C and annealed at constant length at 210° C), including one taken by P. Markey with monochromatic Cu $K\alpha$ radiation, showed twenty-one

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additional reflexions which also fit this cell (cell 2), the dimensions of which are: $a = 4{\cdot}56$ Å, $b = 5{\cdot}94$ Å, $c = 10{\cdot}75$ Å, $\alpha = 98\frac{1}{2}°$, $\beta = 118°$, $\gamma = 112°$ (figure 1). The length of the c axis, which can be found only approximately from normal fibre photographs, was determined more accurately from photographs taken when the fibre axis was oscillated with respect to the X-ray beam over an angular range appropriate for reflexions from crystal planes nearly normal to the fibre axis. Spacings of these planes ($\bar{1}05$, $\bar{2}\bar{1}9$), corrected for the angles of inclination, yielded the value $c = 10{\cdot}75$ Å. In taking oscillation photographs of this type, it is necessary, on account of the difficulty of centring the specimen in a position at right angles to the goniometer axis, to make four exposures on the same film, exposing for one-quarter of the time at each of the following inclinations: (1) at average angle θ, (2) at $\theta + 180°$, (3) at $-\theta$, (4) at $-\theta + 180°$; any error due to eccentricity of the part of the specimen in the X-ray beam is minimized in this way.

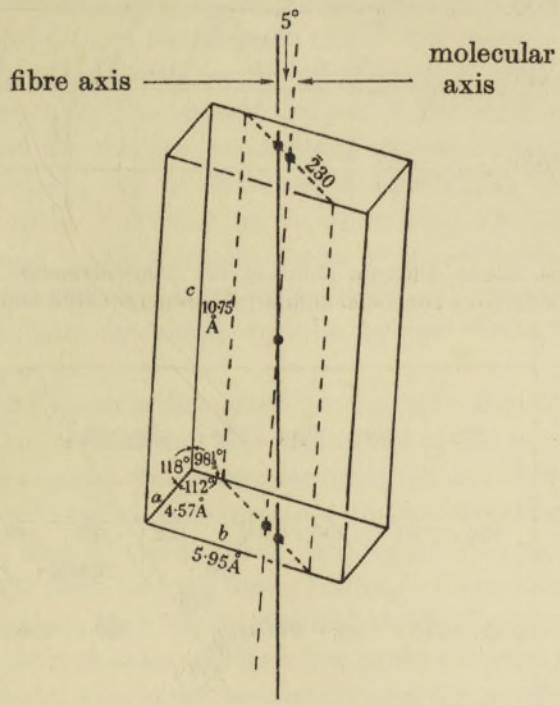


FIGURE 1. Unit cell and direction of tilt in relaxed specimens.

It was found possible to check the unit cell in a very convincing way by making use of diffraction photographs of fibres in which the crystals are tilted in a definite way with respect to the fibre axis. In this type of diffraction pattern, obtained from fibres annealed at 210° C and allowed to contract freely, the reflexions are displaced, some up and some down, by varying amounts from the normal layer-line positions, showing not only that the molecular chain axes of the crystals are tilted with respect to the fibre axis, but further that they are all tilted by about the same amount and all in the same direction with respect to the unit cell. The pattern, illustrated in diagram form in figure 2, is in fact precisely similar to that given by a single crystal rotated around a direction inclined at a few degrees to

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a principal axis, and can be analyzed by the method which has been developed for such patterns (Bunn, Peiser & Turner-Jones 1944). The method, as applied to polyethylene terephthalate, is illustrated in figure 3, which represents the reciprocal lattice net for the third layer line. The origin of the reciprocal lattice is O_3, which must be imagined as being below the plane of the paper. If we imagine the lattice

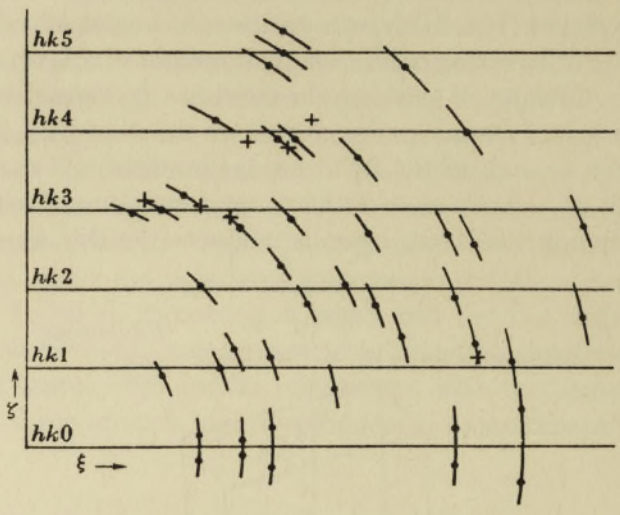


FIGURE 2. Reciprocal lattice diagram showing tilt displacements. Lines show observed position of reflexions, dots show calculated positions for cell 2 and crosses for cell 1.

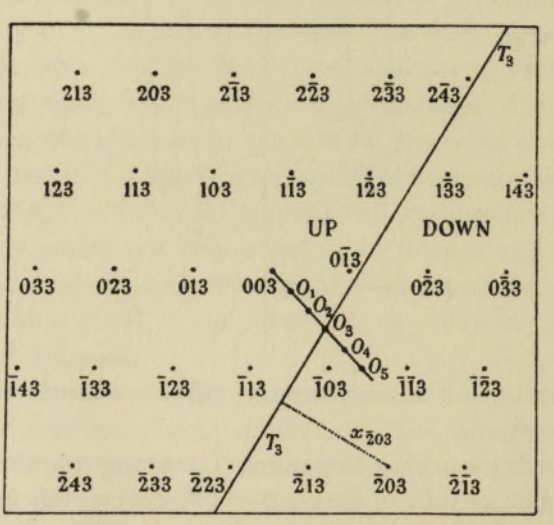


FIGURE 3. Reciprocal lattice net for c projection, illustrating calculation of tilt displacements (third layer).

tilting about a line through O_3 in the basal plane, some of the points on the third layer would move downwards below the paper and others upwards. The diffraction pattern shows that reflexion $0\bar{1}3$ is practically undisplaced from its normal position; hence reciprocal lattice point $0\bar{1}3$ does not move appreciably up or down, and therefore the lattice must be tilted about a line T_3T_3 almost directly under-

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neath point $0\bar{1}3$ and passing through O_3. It is further observed that $\bar{1}\bar{1}3$ is displaced downwards and 003 upwards, hence the lattice must be tilted over to the right. If the lattice, and the indices assigned to the reflexions, are correct, then all $hk3$ reflexions represented by points to the right of T_3T_3 should be displaced downwards and the rest upwards; moreover, the height of any point above the equator (reciprocal lattice co-ordinate ζ) is given by $\zeta = z\cos\phi - x\sin\phi$, where z is the normal untilted height of the reciprocal lattice layer, ϕ is the angle of tilt, and x is the measured distance in the diagram of the point from the tilt line T_3T_3. The same diagram can be used for other layers if alternative origins O_1, O_2, etc., are used; the tilt line for any layer is parallel to T_3T_3 and passes through the origin for that layer. If ϕ is calculated for one reflexion, values of ζ can be calculated for all the others, and these should agree with those found from the photograph. If the reciprocal lattice is incorrect, the calculated tilt displacements will be quite discordant with the observed values in magnitude and sometimes incorrect in sign, and no adjustments of ϕ or of the position of TT will make them correct.

The results of tests of both cells are shown in figure 2. For cell 1, the first three reflexions on the first layer line showed consistent tilt displacements, and these were used to fix ϕ and the tilt line; but some of the other reflexions showed quite incorrect displacements; a few of the worst are shown in figure 2, these being enough to condemn cell 1. For cell 2, on the other hand, all the tilt displacements are of correct sign and within tolerable limits of experimental error.* Unit-cell dimensions checked in this way have the validity and certainty associated with those determined directly for single crystals by the Weissenberg moving film goniometer.

The direction of tilt, which is illustrated for the real cell in figure 1, is such that the $\bar{2}30$ plane remains vertical and the inclination of the 001 plane to the fibre axis increases. The reason for this particular preferred direction of tilt is not clear. That the crystals should tilt in some direction during relaxation is to be expected, because crystals in a fibre are, so to speak, tied together by molecules passing out of one crystal into the next, and the forces holding a tie molecule to neighbouring molecules in a crystal are, for a triclinic crystal, not uniform round the molecular axis; the pull of the tie molecules will therefore cause a crystal to set itself in such a way that its molecular axis is not necessarily exactly parallel to the fibre axis. The actual preferred direction of tilt is presumably determined by the resultant of the holding forces, which is a function of the molecular structure. Examination of the structure described later does not reveal any marked feature associated with the $\bar{2}30$ plane, and it must therefore be concluded that the tilt direction cannot be interpreted in any simple way; it is a complex resultant of molecular forces. The angle of tilt, for the specimen used for this test, was 5°.

Tilted crystal photographs conferred a further advantage: for some reflexions, the ambiguity of indexing inevitable for an untilted specimen, owing to the close similarity of spacing of two or more reflexions on the same layer line, was resolved by the magnitude of the tilt displacements. In this way it was found possible to

* The partial success of cell 1 was due to the fact that it corresponds to the true cell on one projection [001].

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assign unambiguous indices to the great majority of the reflexions; this is seldom possible for high-polymer diffraction patterns, especially when the crystal symmetry is low and the number of different reflexions large. When it is possible, it opens the way to the use of the direct Fourier synthesis method of refinement of atomic parameters. The value of tilted crystal orientation is so great that in any attempt to determine the crystal structure of a high polymer, it would be worth while to try to prepare specimens showing this type of orientation.

The relative intensities of the reflexions were estimated by taking a series of photographs of the same tilted crystal specimen in which the exposures were in the ratios 1:2:4:8:16:32:80; the current through the X-ray tube was kept as steady as possible throughout. Intensities were estimated by visual comparison of the reflexions on the different photographs; some attempt was made to allow for the different sizes of the various reflexions.

Density of crystalline regions and proportion of crystals in specimens

The unit cell contains only one chemical unit (—O . CH_2 . CH_2 . O . CO . C_6H_4 . CO—). From the unit-cell dimensions, the volume of the cell is 219 Å^3; the absolute weight of one chemical unit is 319×10^{-24} g, hence the density of the crystal is 1·455. The usual density of drawn yarn is about 1·39, while the highest density found for a crystalline specimen is 1·41; these values are lower than the crystal density owing to the presence of amorphous material. If it is assumed that the density of the amorphous regions in a partly crystalline specimen is the same as that of entirely amorphous material (1·335), the proportion by weight of crystalline regions in ordinary drawn yarn is found to be 48 %, and in specimens of the highest recorded density 64·5 %. These figures, however, are lower limits; if, as is likely, the density of amorphous regions in partly crystalline specimens is lower than 1·335 (see Matthews, Peiser & Richard 1949; Bunn 1953), the proportions of crystalline material in the specimens are higher than the figures given.

Determination of chain configuration and atomic positions

Since a flat molecule would have a repeat distance of 10·9 Å, and the measured value is 10·75 Å, it is evident that the molecule is nearly planar; a small deviation from planarity would shorten the repeat distance by the required amount. The great intensity of the 100 reflexion suggests that the plane of the molecule is roughly parallel to the (100) plane. Starting from this point, the relative intensities which would be given by a range of molecular configurations in various orientations in the unit cell were calculated and compared with the observed intensities. In considering non-planar configurations, it was assumed that the molecule retains a centre of symmetry—i.e. the space group of the crystal is $P\bar{1}$; although it is possible to make non-centrosymmetric models (for instance, by rotating round the two O—CH_2 bonds by different amounts), it seems very unlikely that such configurations could exist (except momentarily in the course of thermal vibrations) in this crystal structure, for the consideration of models suggests that in a triclinic cell containing only one molecule which is potentially centrosymmetric, inter-

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molecular forces would keep the mean configuration centrosymmetric. It was expected that the departure from planarity is most likely to arise from rotations round the O—CH_2 bonds (in such a way as to preserve centrosymmetry), since this has been found to occur in the structure of the aliphatic polyester, polyethylene adipate $(—O(CH_2)_2O.CO(CH_2)_4CO—)_p$ (Bunn & Turner-Jones unpublished).

A fair degree of agreement between calculated and observed intensities was attained for one set of atomic positions, but great difficulty was experienced in achieving further refinement by the trial and error method. The Fourier synthesis method was then invoked; fortunately, this was possible because nearly all the reflexions had been unambiguously indexed by the tilted crystal method. Only three-dimensional electron density computations could be expected to be useful, the data from any one projection being insufficient to yield useful results. To keep

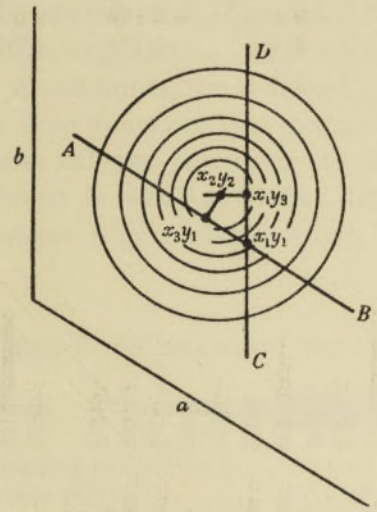


FIGURE 4. Determination of co-ordinates from electron density distribution along lines.

the labour of computation within reasonable limits, electron densities were calculated only in the vicinity of atomic centres, and only at points along lines parallel to unit-cell edges and passing through the approximate atomic positions already found. The procedure is illustrated two-dimensionally in figure 4: suppose the approximate co-ordinates of an atom (obtained by trial) are x_1y_1 but the position of maximum electron density for the set of terms used in the synthesis is really at x_2y_2; calculations of electron densities along the lines AB and CD reveal maxima at x_1y_3 and x_3y_1; perpendiculars on these lines at these points lead to the correct position of the maximum x_2y_2, assuming that the electron density contours are circular. The computations, which were done with the aid of Beevers–Lipson strips, lead to atomic co-ordinates which gave improved agreement between calculated and observed intensities; the process was repeated, with inclusion of more and more higher-order terms, until all the observed reflexions were included and no further changes of sign took place. It was found that not all the atoms were sufficiently well resolved to give reliable atomic co-ordinates in this way;

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oxygen atoms which have a considerably higher electron density than carbon atoms, tended to smudge over the carbon atoms attached to them, so that the peaks of the latter were rather ill-defined. Some final adjustments of the positions of these particular atoms by trial-and-error calculations were therefore made. The final atomic co-ordinates, in fractions of unit cell edge lengths and in ångströms, are given in table 1.

TABLE 1. ATOMIC CO-ORDINATES

	atom	x	y	z	x	y	z
1	C	$0{\cdot}000a$	$-0{\cdot}078b$	$+0{\cdot}371c$	0·00 Å	−0·47 Å	+3·99 Å
2	C	0·000	−0·181	+0·237	0·00	−1·08	+2·55
3	C (of CH)	+0·100	+0·170	+0·438	+0·46	+1·01	+4·71
4	C (of CH)	−0·117	−0·257	+0·433	−0·53	−1·53	+4·66
5	C (of CH_2)	+0·045	−0·085	+0·040	+0·21	−0·51	+0·43
6	O (of C=O)	−0·098	−0·415	+0·170	−0·45	−2·47	+1·83
7	O	+0·052	0	+0·175	+0·24	0·00	+1·88

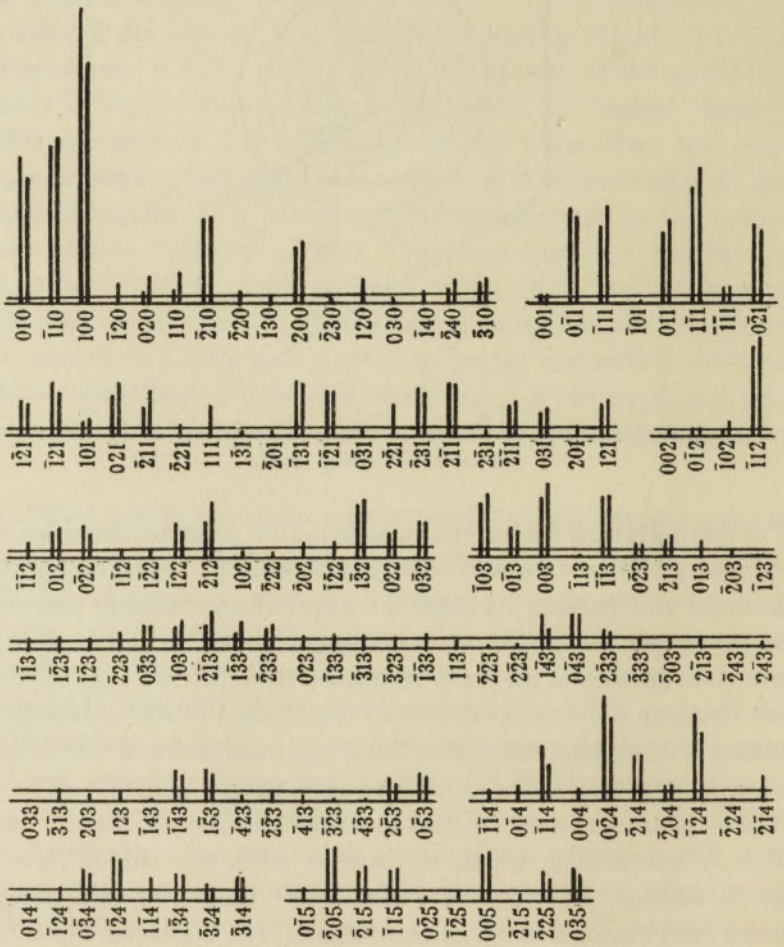


FIGURE 5. Comparison of observed and calculated structure factors. In each pair of lines, the length of the one on the left is proportional to the observed structure factor, that on the right to the calculated value. Where only one line is shown it represents calculated intensity. The line just above the base-line indicates the limit of visibility.

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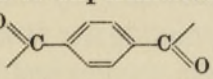
The structure factors calculated for these atomic positions, and the observed values, are compared in figure 5. For these final calculations, the scattering powers of C, CH, CH_2 and O were assumed to be in the ratios 6:7:8:10. A temperature factor $\exp -B(\sin\theta/\lambda)^2$ with $B = 9\times10^{-16}$ was used, this giving the correct rate of fade-off with increasing angle of diffraction θ; the fade-off is probably due partly to thermal motions of the molecules and partly to crystal imperfections. The so-called 'reliability factor' $\frac{\Sigma\,|F_{\text{calc.}}-F_{\text{obs.}}|}{\Sigma\,|F_{\text{obs.}}|}$ for these figures is 0·19, which for a high polymer structure is considered very satisfactory.

Description of structure and interatomic distances

Within the molecule the distances between bonded atoms agree fairly well with these found in the closely related molecule diethyl terephthalate (Bailey 1949) (table 2). Between atoms in neighbouring molecules all the distances are those characteristic of van der Waals forces, the shortest in the crystal being C—C 3·80, 3·95 and O—C 3·50. There is no indication of abnormally short distances, and thus no indication of strong forces between particular atoms of neighbouring molecules. This conclusion is significant in relation to the physical properties; the high melting-point is evidently not due to strong forces between the molecules like those in the polyamides.

Table 2. Interatomic distances

	polyethylene terephthalate	diethyl terephthalate
1–2 (C of benzene ring to C of C=O)	1·49	1·48
1–3 (C—C, benzene ring)	1·34	1·38
1–4 (C—C, benzene ring)	1·36	1·40
2–6 (C=O)	1·27	1·28
2–7 (C—O)	1·34	1·32
5–7 (C—O)	1·44	1·51
5*a*–5*b* (C—C, aliphatic)	1·49	1·55

The configuration of the molecule is illustrated in figure 6. Carbon and oxygen atoms lie nearly but not quite in a plane. The  group appears to be about 12° out of the plane of the benzene ring; to what extent this apparent departure from planarity is due to experimental error is not certain; one would expect the conjugated system  to be substantially planar, though small departures from planarity are not impossible. A larger departure from planarity occurs in the glycol group; it can be described as a rotation of the CH_2—CH_2 bond (round the O—CH_2 bond as axis) of about 20° from the planar configuration. This departure from planarity is of the same type as that found in the aliphatic polyester, polyethylene adipate, but in the latter the rotation (80°) is much greater (Bunn & Turner-Jones, unpublished). The difference between the ester

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group configurations in polyethylene terephthalate and polyethylene adipate suggests that rotation round the O—CH_2 bond requires very little energy, and that the departure from planarity is less in polyethylene terephthalate simply because a more nearly planar molecule fits better with its neighbours than a very non-planar type.

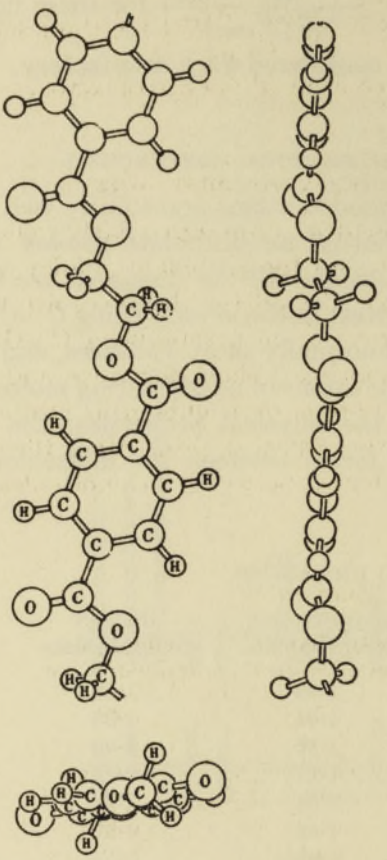


FIGURE 6. Configuration of molecule.

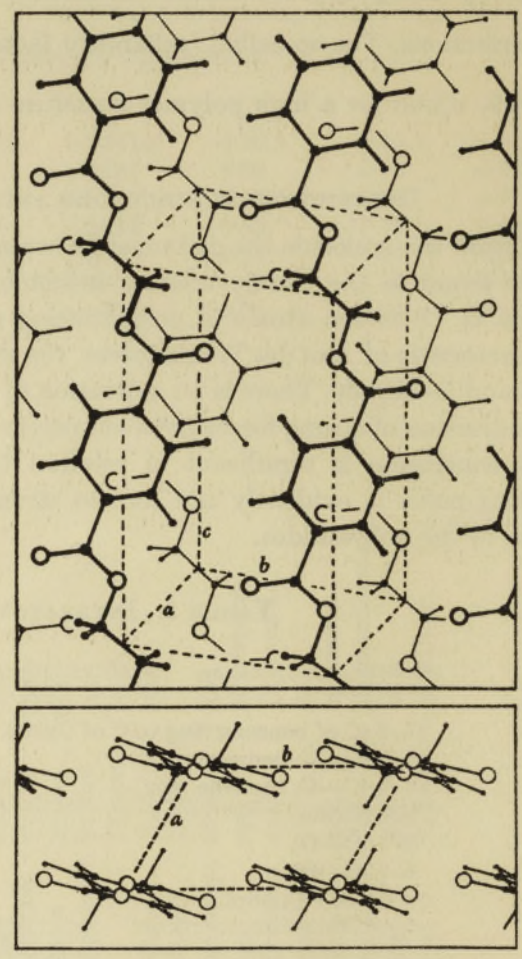


FIGURE 7. Arrangement of molecules in crystal. Above, projection normal to 010 plane; below, projection along c axis. (Larger dots, carbon; smaller dots, hydrogen; open circles, oxygen.)

The mode of packing of the molecules (see figure 7) appears to be chiefly a matter of shape; if it is assumed that carbon, oxygen and hydrogen atoms have effectively spherical domains with radii 1·7, 1·35 and 1·2 Å respectively, the molecules fit in with each other fairly well—projections on one molecule fit into hollows of the next, and there are no large unfilled spaces. In view of what was said in the previous paragraph, it should be possible to explain the precise configuration of the glycol part of the molecule on the basis of packing considerations; but this does not appear possible at present, probably because our conceptions of

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effective shape are too crude; the assumption of effectively spherical shape of atoms is not an entirely satisfactory substitute for the realities of van der Waals forces.

The melting-point

One of the principal reasons for the value of polyethylene terephthalate as a textile fibre is its high melting-point, 264° C, which is very close to that of nylon 66 (265° C). High melting-points are characteristic of aromatic polyesters (Hill & Walker 1948), which contrast sharply with aliphatic polyesters, the melting-points of which are all well below 100° C. The melting-points of chain polymers will be discussed in a more general context elsewhere; here it will be of interest to say something about polyethylene terephthalate itself. The principal additional facts to be taken into account are that unbranched polymethylene $(—CH_2—)_p$, which

TABLE 3. COHESION ENERGIES IN POLYMER CRYSTALS

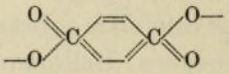
	molar c.e. (cal) per chain atom	cal/cm chain length	cal/cm² mol. surface area	cal/cm³	m.p. (°C)
polymethylene $(—CH_2—)_p$	1300	1600×10^{-16}	108×10^{-8}	90	130
polyethylene adipate $(—O(CH_2)_2O—CO(CH_2)_4CO—)_p$	1800	2500	165	139	52
polyethylene suberate $(—O(CH_2)_2O—CO(CH_2)_6CO—)_p$	1600	2200	147	123	~55
polyethylene terephthalate $(—O(CH_2)_2O—CO.C_6H_4.CO—)_p$	1900	2900	170	143	264
nylon 66 $(—NH(CH_2)_6NH—CO(CH_2)_4CO—)_p$	3400	4600	300	260	265

may be taken as the reference substance, melts at about 130 to 136° C (Kantor & Osthoff 1953, Mandelkern, Hellmann, Brown, Roberts & Quinn 1953), while the two aliphatic polyesters most nearly comparable with polyethylene terephthalate —polyethylene adipate $(—O(CH_2)_2O—CO(CH_2)_4CO—)_p$, which has about the same length of repeating unit, and polyethylene suberate $(—O(CH_2)_2O—CO(CH_2)_6CO—)_p$, which has the same number carbon atoms in the repeating unit—melt at 50 to 60° C (Carothers & Arvin 1929; Fuller & Erickson 1937).

The cohesive forces between polyester molecules are much greater than those between hydrocarbon chains, owing to the dipoles in the ester groups. These forces can be estimated approximately by an additive scheme in which each functional group of atoms is associated with a definite increment of cohesion energy (Bunn 1953). The scheme works well for monomeric substances where the results can be checked against the measured latent heats of vaporization, and it can therefore be applied confidently to polymers. The figures for the polymers under consideration (corrected for crystal density) are given in table 3; since there may be some difference of opinion about the basis of comparison, the results are expressed in four different ways: (i) as calories per chain atom (molar units), (ii) as calories per centimetre length of chain, (iii) as calories per square centimetre surface area of the molecule

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(which is treated as a rod of elliptical cross-section, the dimensions being given by the crystal structure), and (iv) as calories per cubic centimetre (the last being the cohesive energy density, a quantity often used in discussing solubility relations). The writers consider that the energy per unit surface area of the molecule is the most appropriate, but whatever basis is favoured, the conclusions are essentially the same. The cohesive forces in the aliphatic polyesters are considerably higher than in the hydrocarbon polymer; yet the melting-points of the polyesters are much lower. There can be little doubt that this is due to enhanced molecular flexibility; the importance of molecular flexibility in determining polymer melting-points has been stressed elsewhere (Bunn, 1942, 1953), and in the polyesters there are reasons for believing that it is the O—CH_2 bonds (not the CO—CH_2 bonds) which confer a flexibility greater than that of a CH_2 chain; one of the reasons is that all chain ethers R_1—O—R_2 as well as chain esters R_1—CO—O—R_2, monomers as well as polymers, melt lower than corresponding hydrocarbons, whereas ketones $R_1CO.R_2$ melt higher; another is the already mentioned difference of configuration between different polyester molecules, suggesting easy rotation round the O—CH_2 bonds.

Polyethylene terephthalate also contains O—CH_2 bonds which permit easy rotation; yet its melting-point is very much higher than the aliphatic polyesters. The cohesive forces are only a little higher than in the aliphatic polyesters; this is in agreement with the conclusion from the crystal structure that there are no strong forces between individual atoms in neighbouring molecules, and also shows that the more general attractions associated with aromatic rings (which do not show up in interatomic distances) are only a little greater than in aliphatic compounds. The small difference cannot explain the high melting-point, since the cohesion energy figures for nylon 66 (which melts at the same temperature) are very much higher. The higher melting-point is evidently due to the stiffening of the aromatic part of the chain; the whole of the group  would be expected to be rigid owing to the conjugated system of double bonds, and the stiffening of these parts of the chain evidently more than outweighs the opposite effect of the O—CH_2 bonds. Similar relations are found between comparable monomeric esters: diethyl adipate melts at $-21°$ C, while diethyl terephthalate melts at 44° C.

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